June 28, 2023

Meet at **noon** tomorrow!

12pm-1pm : **Lecture**

1pm-2pm : **Studio**

Reflection 1 : Due Friday by midnight

Conceptual HW 1 : Due Sunday by midnight

Office Hours

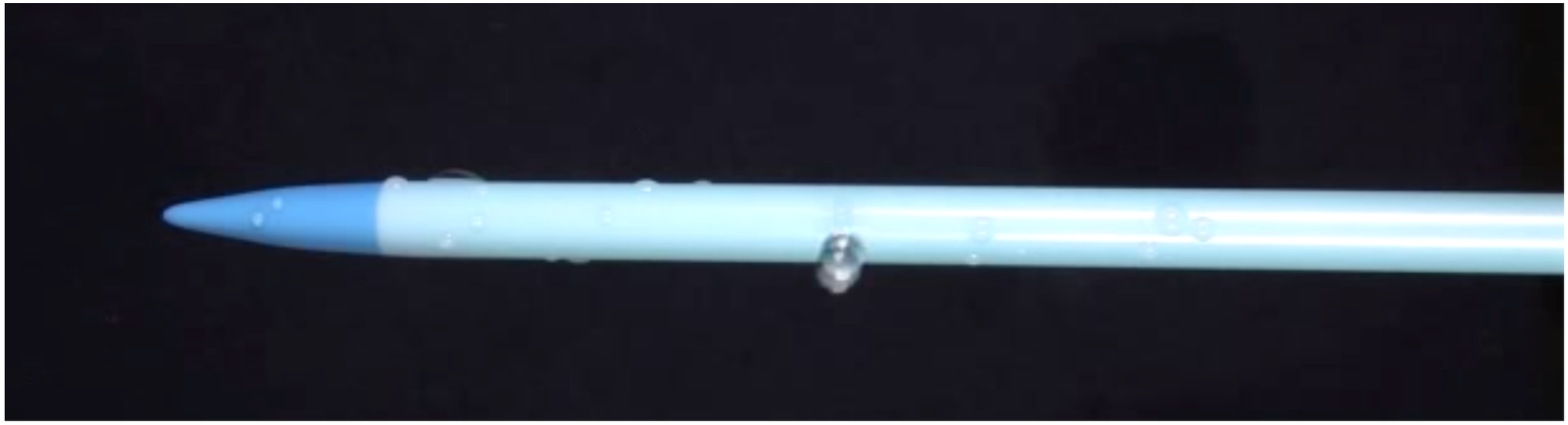
Mondays Noon-2pm

Thursdays 2-3pm

	Monday	Tuesday	Wednesday	Thursday	Friday
Week 1	June 26 Intro, V2 CH5	June 27 Lab W1 - Elec Force	June 28 V2 CH5	June 29 V2 CH5 Studio : Force & Field	June 30 V2 CH5 Reflection
Week 2	July 3 No Class	July 4 No Classes!	July 5 V2 CH 5	July 6 V2 CH 6 Homework 1 Studio : Field Calcs	July 7 V2 CH 6 Reflection
Week 3	July 10 V2 CH 6	July 11 Homework 2 Lab W3 - Using E	July 12 V2 CH 7	July 13 V2 CH 7 Studio : Gauss's Law	July 14 V2 CH 7 Reflection

Chapter 5

Electric Charges and Fields



Science off the Sphere: Knitting Needle Experiment

NASA: Johnson Space Center (Houston)

https://www.youtube.com/watch?v=qHrBhgwq__Q

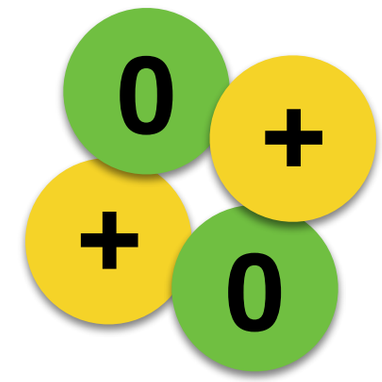
Electric Forces

§ 5.3
Lecture 02

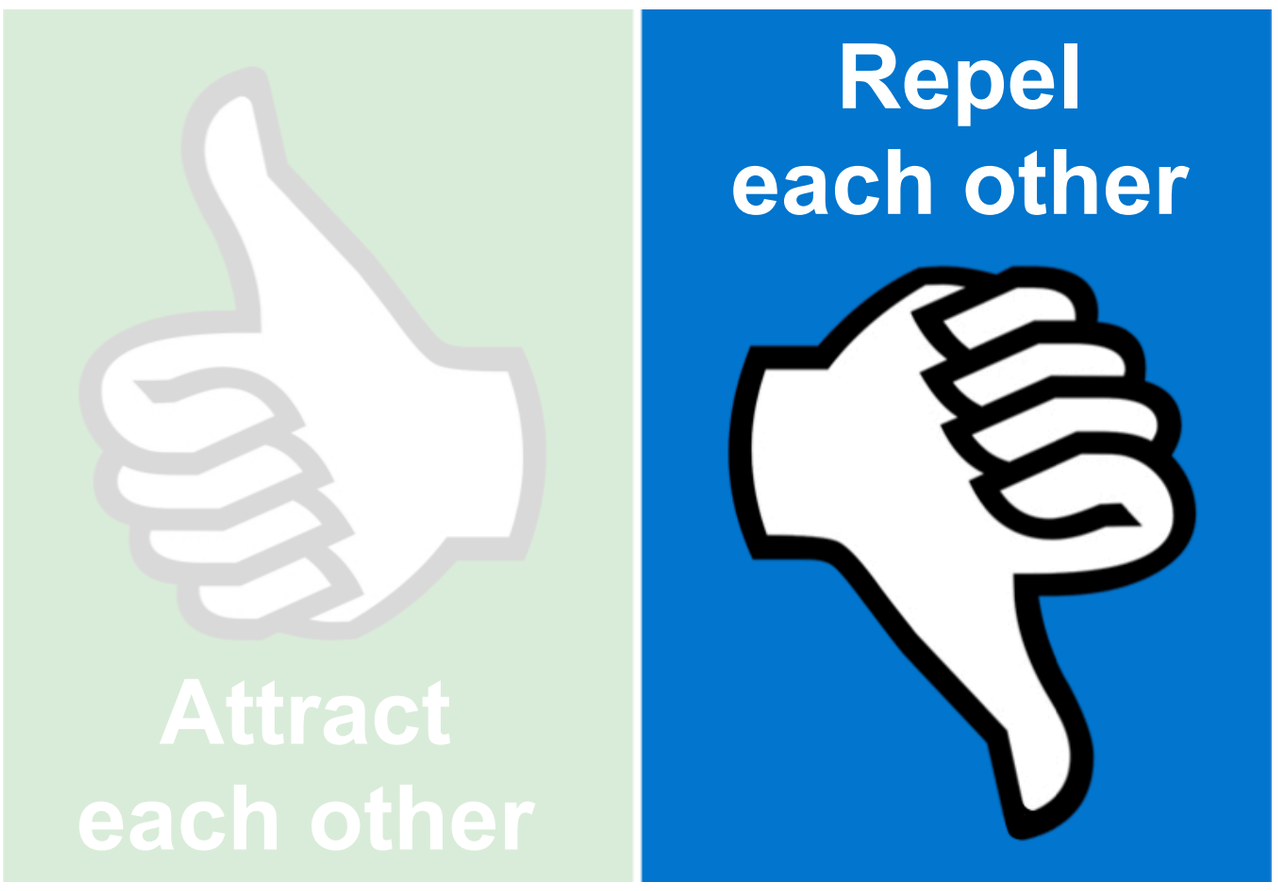
Helium Nucleus

Particle Interaction

A helium nucleus is made of two neutrons (have a charge of 0C) and two protons.



The two protons…

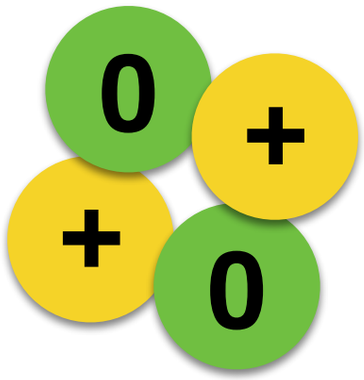


0. Don’t wait on my response.

Helium Nucleus

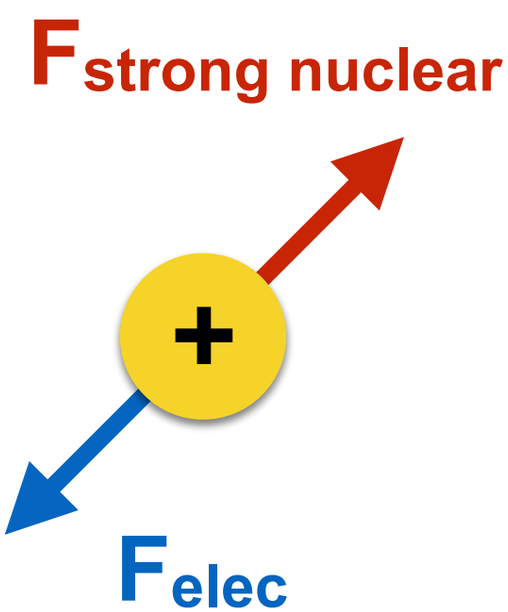
Nucleus Size

A helium nucleus is made of two neutrons and two protons.



The two protons repel each other, but are held together by the **Strong Nuclear Force.** This force has a magnitude of about 40.0N.

Draw the free body diagram for a proton.



Calculate how far apart the protons in a helium nucleus are.

$$|F_{-8}| = K\frac{|q_{-8}|\,|q_{-3}|}{r^2} \qquad 40N = K\frac{|e|\,|e|}{r^2}$$

$$\mathbf{r = 2.40 \times 10^{-15}\ m}$$

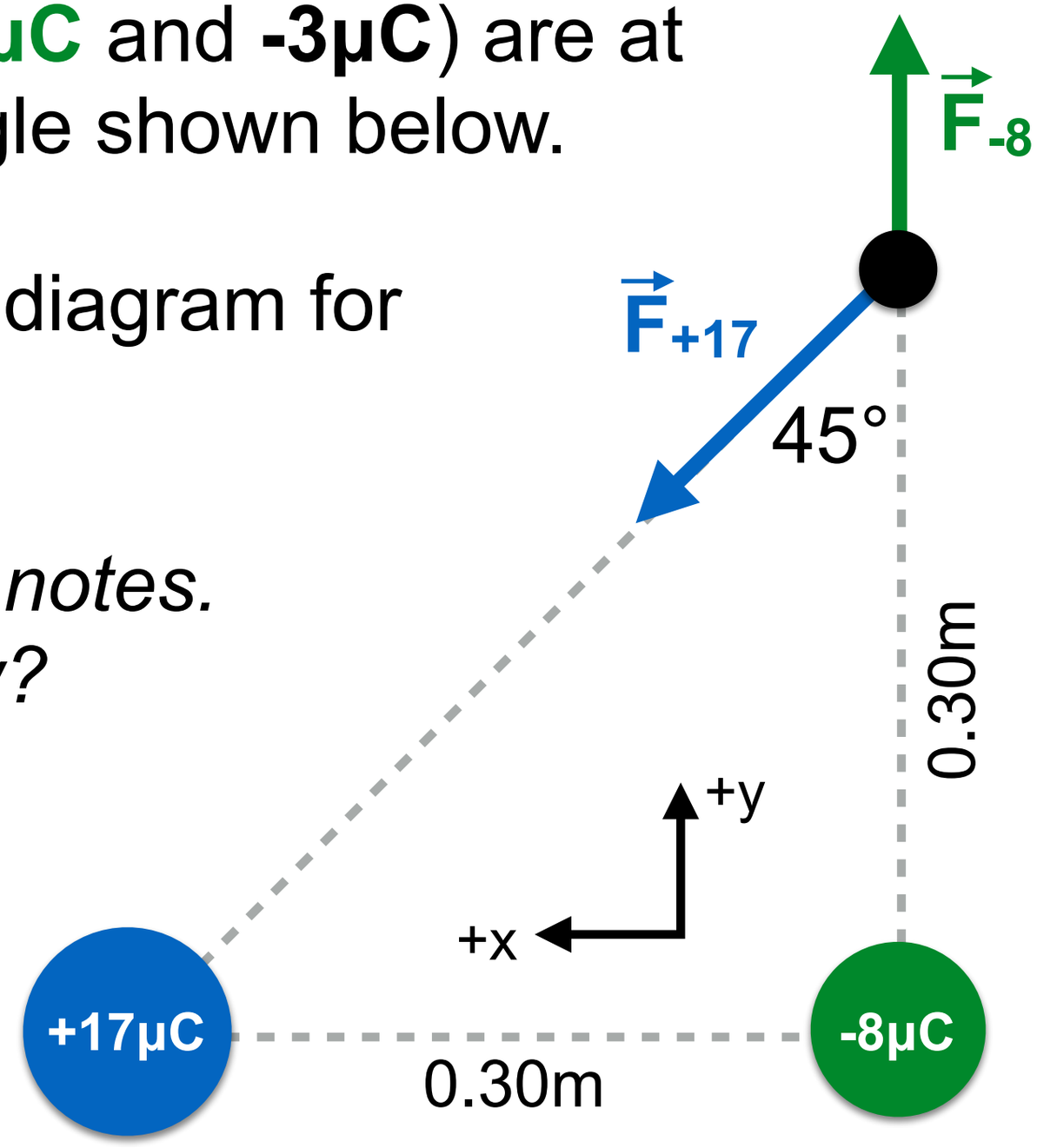
Summing Electric Forces

Direction for each charge

Three charges (**+17μC**, **-8.0μC** and **-3μC**) are at the corners of the right triangle shown below.

Step 1 : Draw the free body diagram for the top charge.

Draw the diagram in your notes.
What angles do you know?



Summing Electric Forces

Force Vectors (- charge)

Three charges (**+17μC**, **-8.0μC** and **-3μC**) are at the corners of the right triangle shown below.

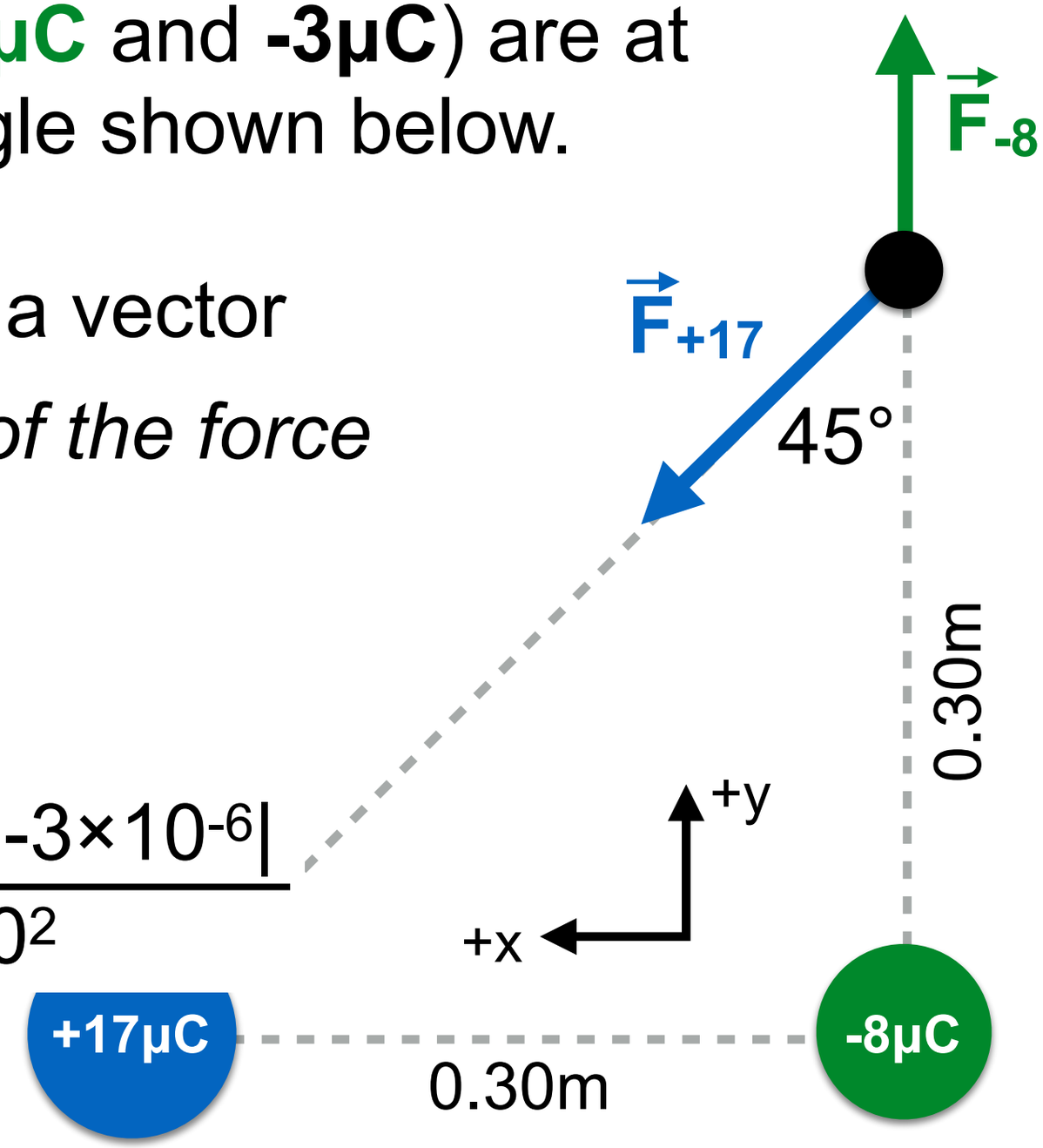
Step 2 : Write each force as a vector

Calculate the magnitude of the force from the -8μC charge.

$$|F_{-8}| = K\frac{|q_{-8}|\,|q_{-3}|}{r^2}$$

$$= (8.99\times10^{9})\,\frac{|-8\times10^{-6}|\,|-3\times10^{-6}|}{0.30^2}$$

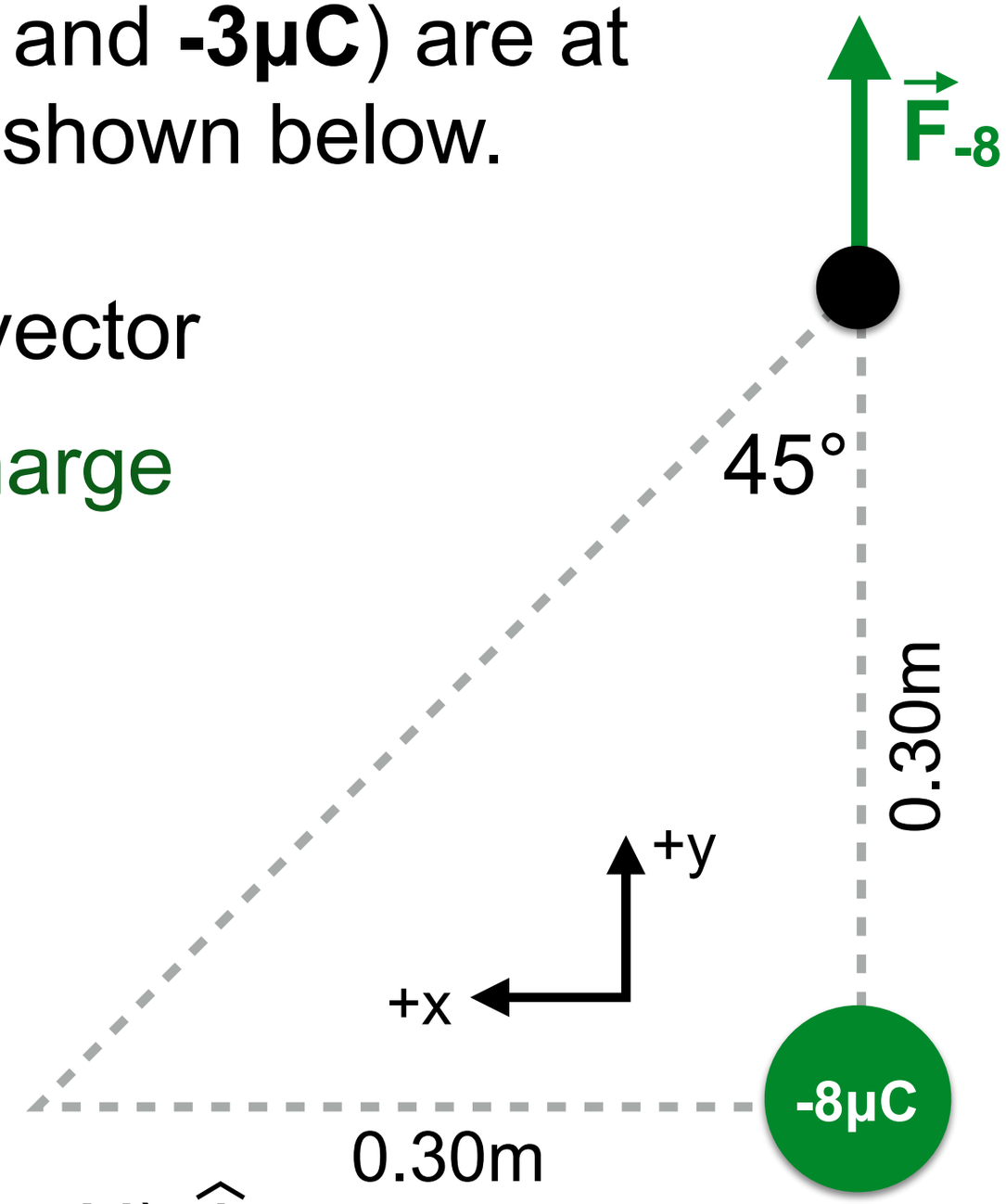
$$|F_{-8}| = 2.4\ \text{N}$$



Summing Electric Forces

Force Vectors (- charge)

Three charges (**+17μC**, **-8.0μC** and **-3μC**) are at the corners of the right triangle shown below.



Step 2 : Write each force as a vector

The force from the -8μC charge *as a vector* is…

1. $\vec{F}_{-8} = (+\ 2.4\ \text{N})\ \hat{i}$
2. $\vec{F}_{-8} = (+\ 2.4\ \text{N})\ \hat{j}$
3. $\vec{F}_{-8} = (-\ 2.4\ \text{N})\ \hat{i}$
4. $\vec{F}_{-8} = (-\ 2.4\ \text{N})\ \hat{j}$
5. $\vec{F}_{-8} = (+\ 1.7\ \text{N})\ \hat{i} + (+\ 1.7\ \text{N})\ \hat{j}$

Summing Electric Forces

Force Magnitudes (+ charge)

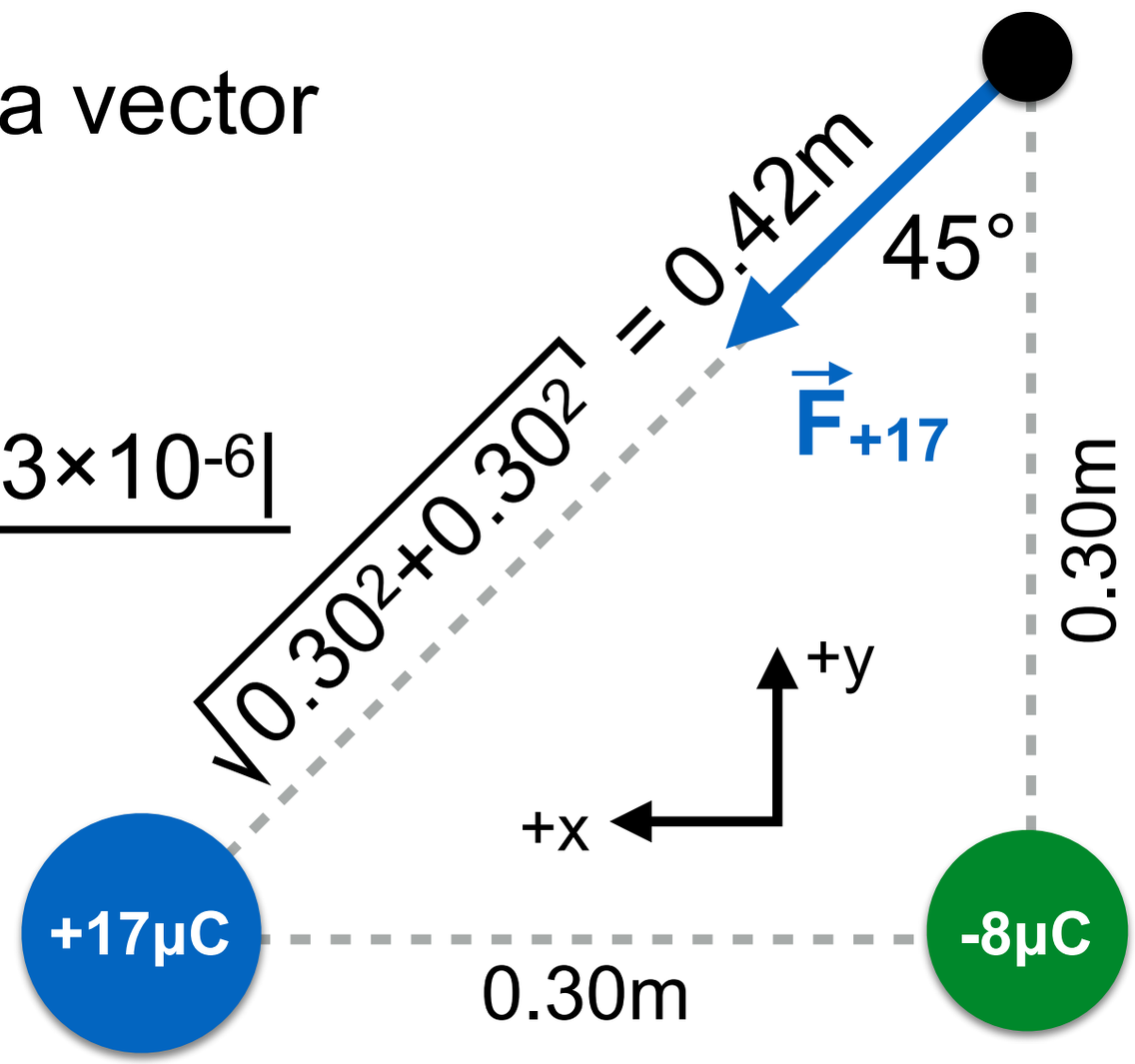
Three charges (**+17μC**, **-8.0μC** and **-3μC**) are at the corners of the right triangle shown below.

Step 2 : Write each force as a vector

$$|F_{+17}| = K\frac{|q_{+17}|\,|q_{-3}|}{r^2}$$

$$= (8.99\times10^{9})\frac{|17\times10^{-6}|\,|-3\times10^{-6}|}{0.42^2}$$

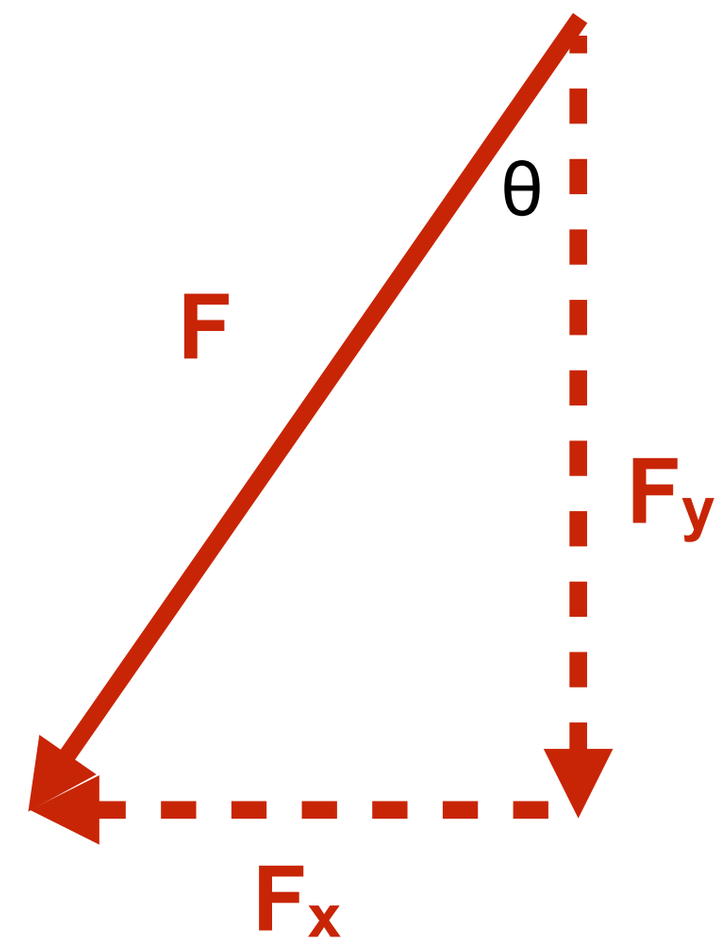
$$|F_{+17}| = 2.6\text{ N}$$



Trigonometry Review

Calculating Sides

For the electric force vector shown, how is its x-component related to the magnitude and the angle θ?



1. $F_x = F \sin(\theta)$
2. $F_x = F \cos(\theta)$
3. $F_x = F \tan(\theta)$
4. $F_x = F / \sin(\theta)$
5. $F_x = F / \cos(\theta)$
0. Don't wait on my response

Summing Electric Forces

Force Magnitudes (+ charge)

Three charges (**+17μC**, **-8.0μC** and **-3μC**) are at the corners of the right triangle shown below.

Step 2 : Write each force as a vector

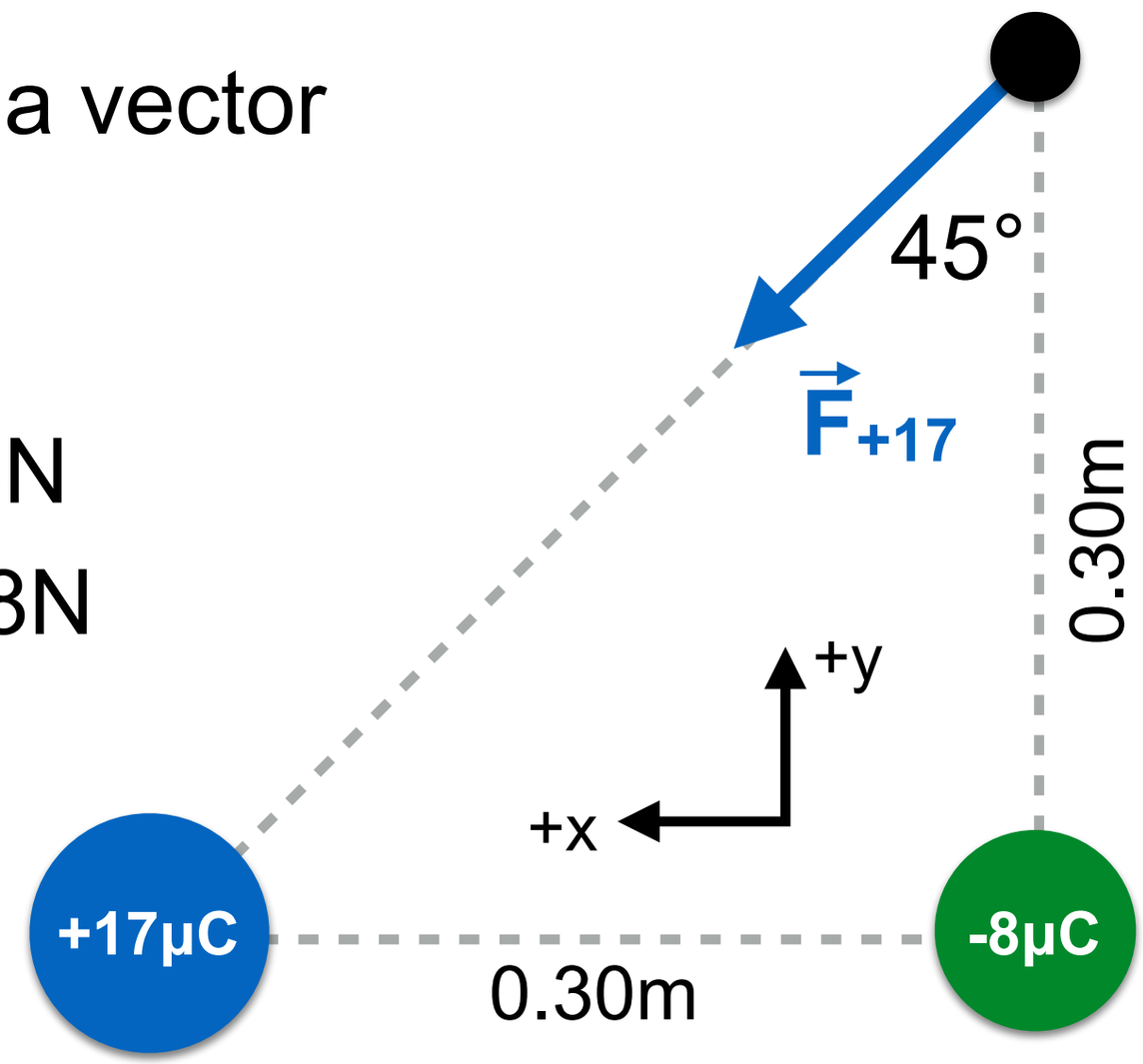
$|F_{+17}| = 2.6$ N

$F_{+17,x} = (2.6\text{ N})\sin(45°) = 1.8\text{N}$

$F_{+17,y} = (2.6\text{ N})\cos(45°) = 1.8\text{N}$

$\vec{F}_{+17} = (+\ 1.8\text{ N})\ \hat{i} +$

$(-\ 1.8\text{ N})\ \hat{j}$



Summing Electric Forces

Adding vectors : Find Components

Three charges (**+17μC**, **-8.0μC** and **-3μC**) are at the corners of the right triangle shown below.

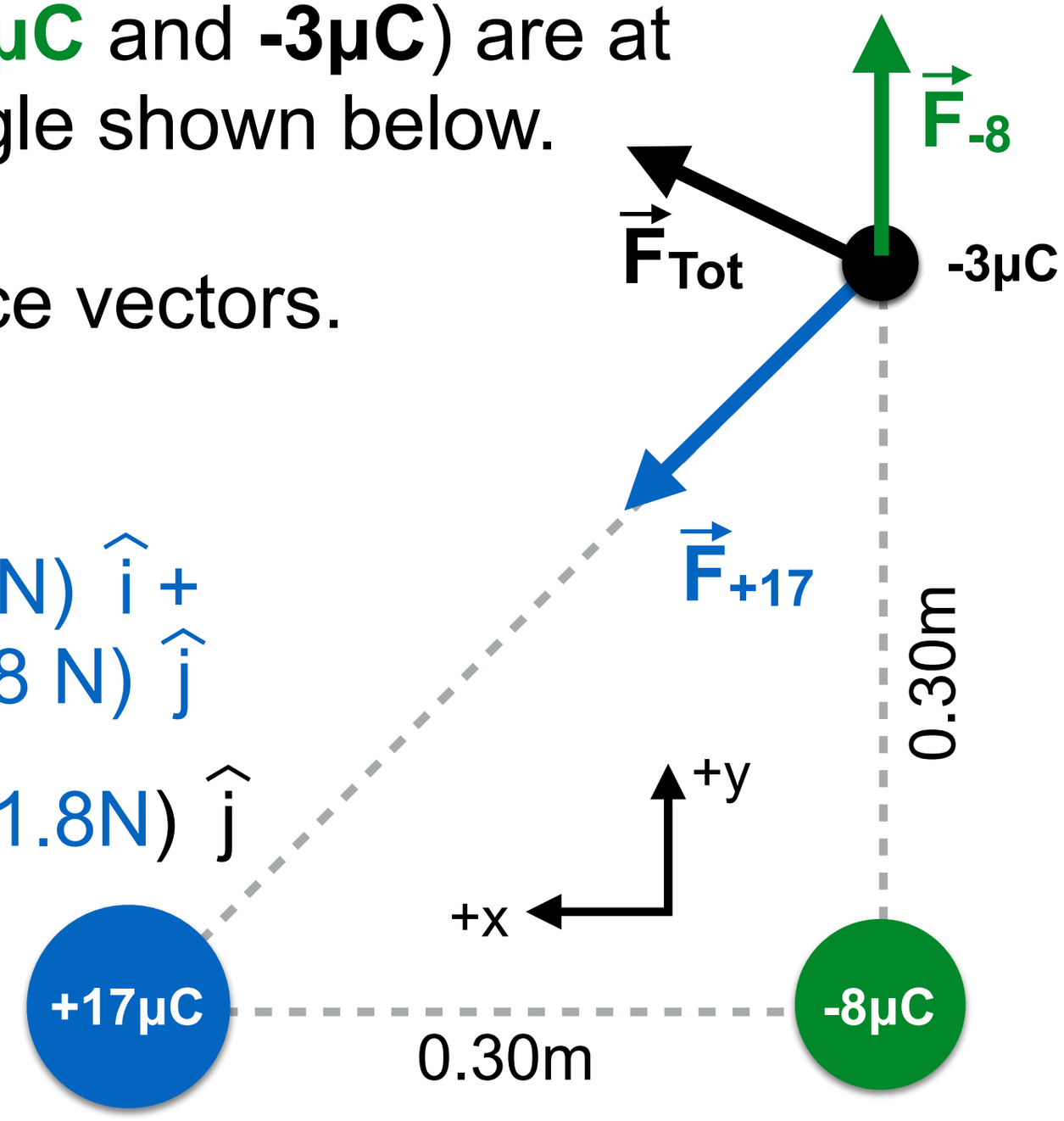
Step 3 : Add together all force vectors.

$$\vec{F}_{tot} = \vec{F}_{-8} + \vec{F}_{+17}$$

$$\vec{F}_{tot} = (+\ 2.4\ N)\ \hat{j} + (+\ 1.8\ N)\ \hat{i} + (-\ 1.8\ N)\ \hat{j}$$

$$\vec{F}_{tot} = (+1.8N)\ \hat{i} + (+2.4N-1.8N)\ \hat{j}$$

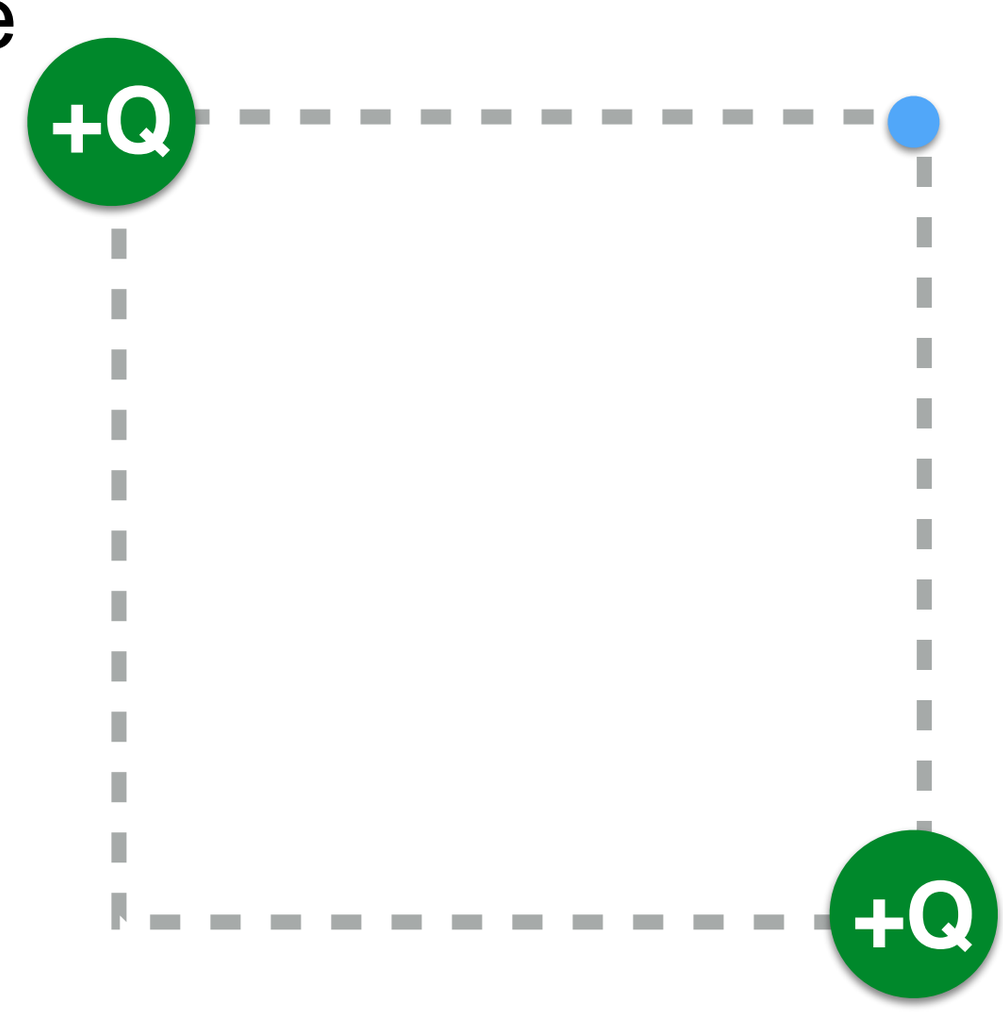
$$\vec{F}_{tot} = 1.8N\ \hat{i} + 0.6N\ \hat{j}$$

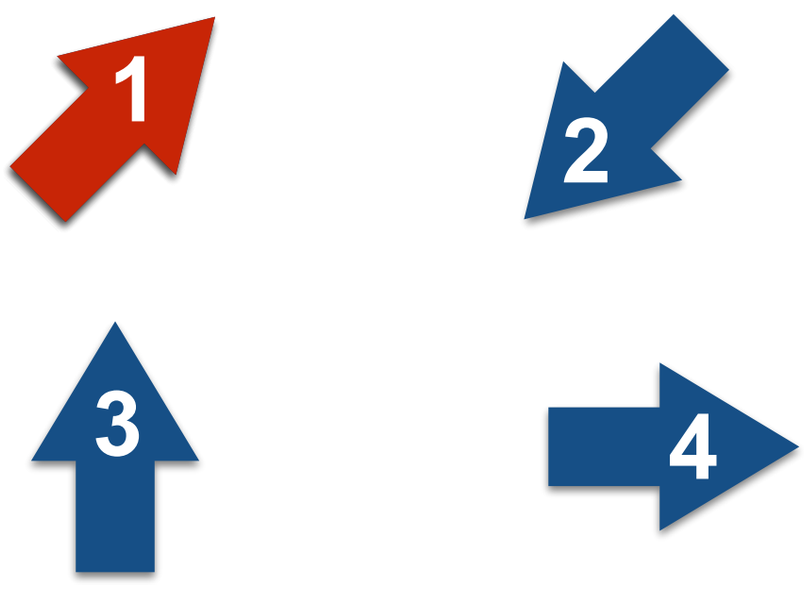


Check Your Understanding

Combined Electric Forces

If there were a +Q charge at the top-right corner of the square, what is the direction of the net force on it?



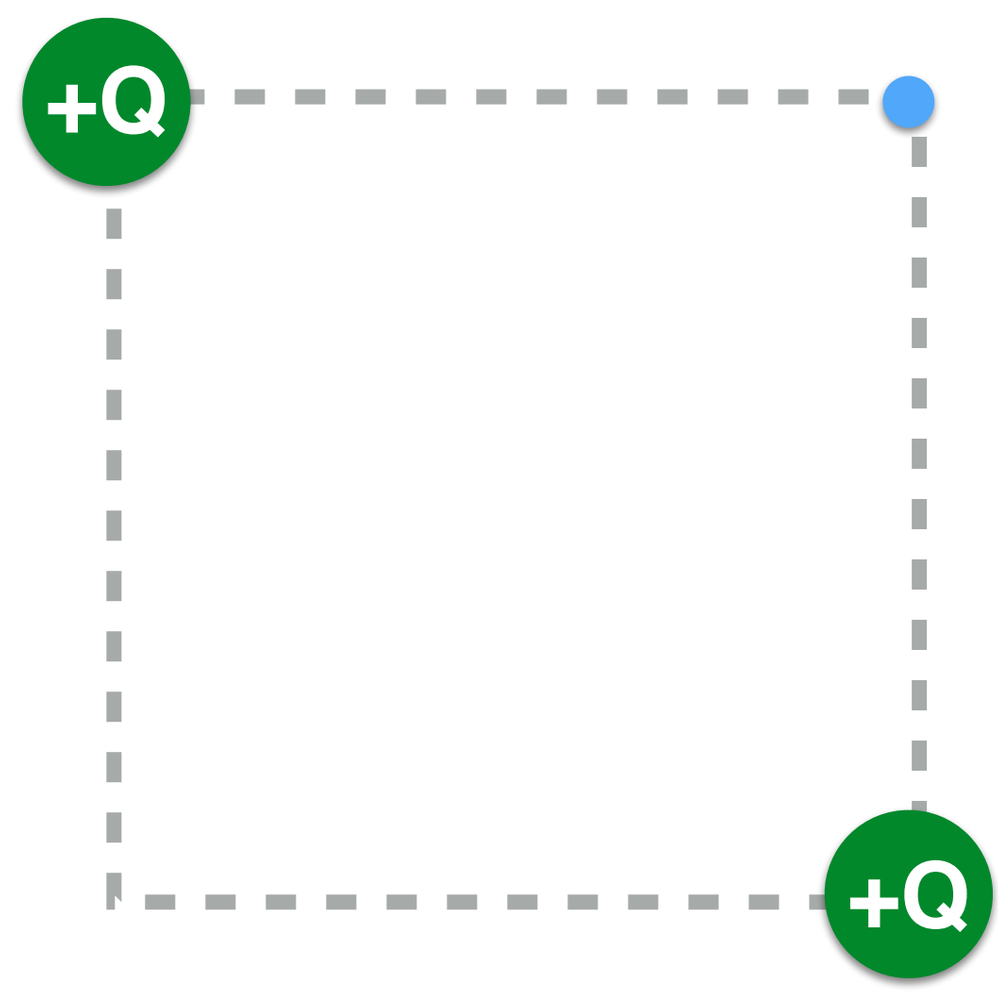


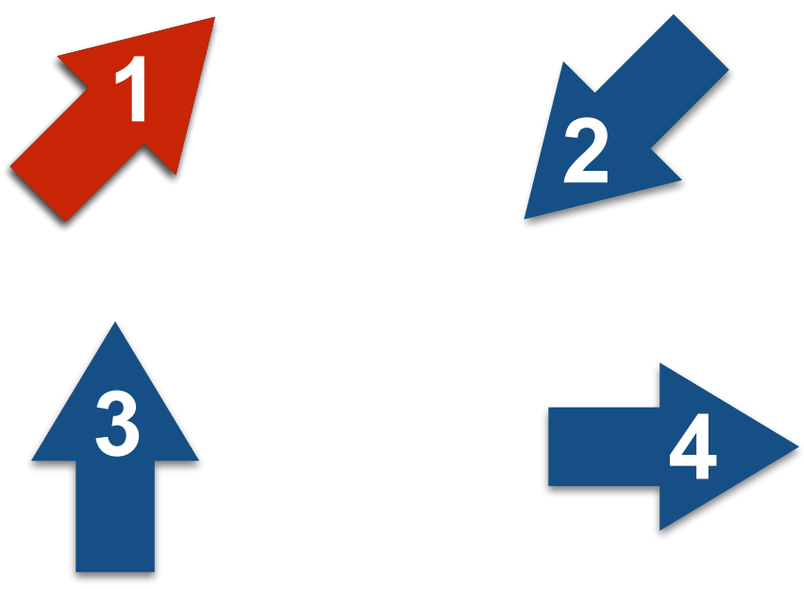
5. The net force is zero
0. Don’t wait on my respone

Check Your Understanding

Combined Electric Forces 2

If there were a +5Q charge at the top-right corner of the square, what is the direction of the net force on it?



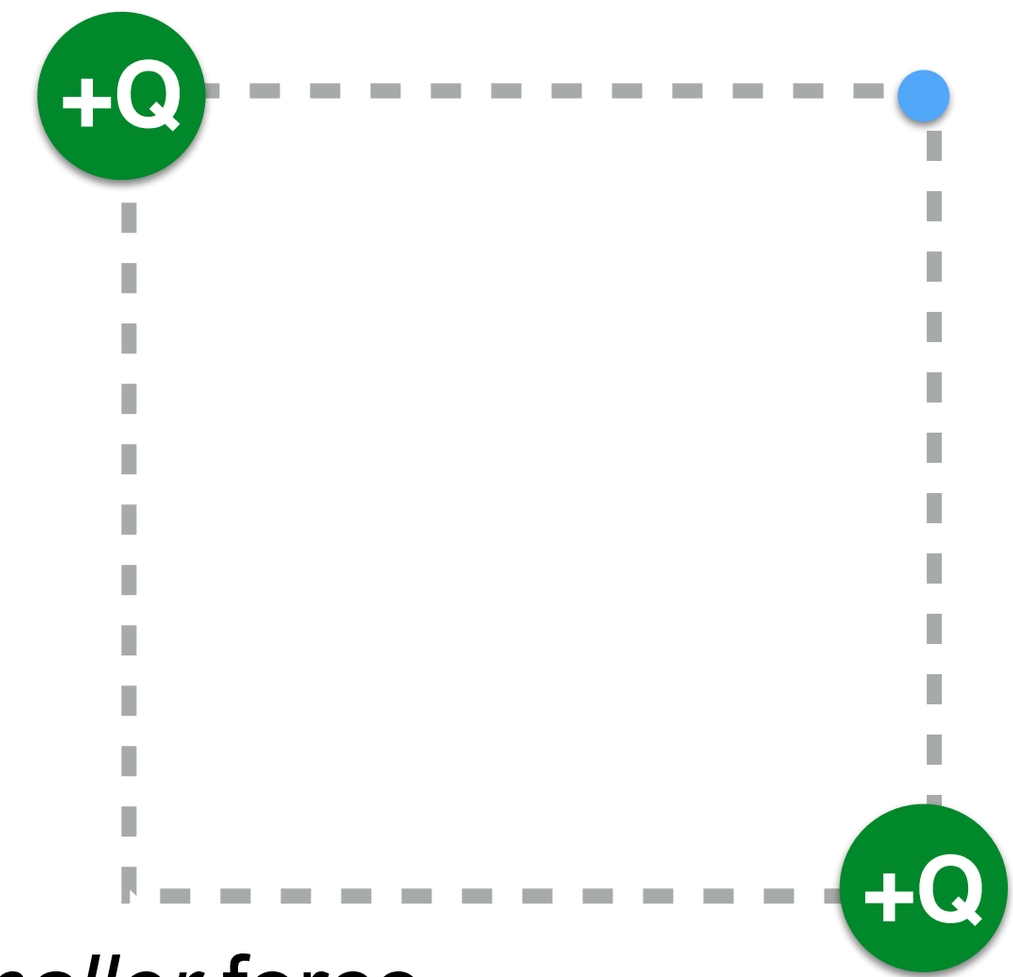


5. The net force is zero
0. Don’t wait on my respone

Check Your Understanding

Combined Electric Forces 3

How many times larger is the force on a charge at the top-right corner of the square if it were a +5Q charge compared to a +Q charge?



1. 5-times larger
2. 2.5-times larger
3. 25-times larger
4. Trick question! It'd be a *smaller* force.
5. None of these are correct
0. Don't wait on my response

Pre-class Slides

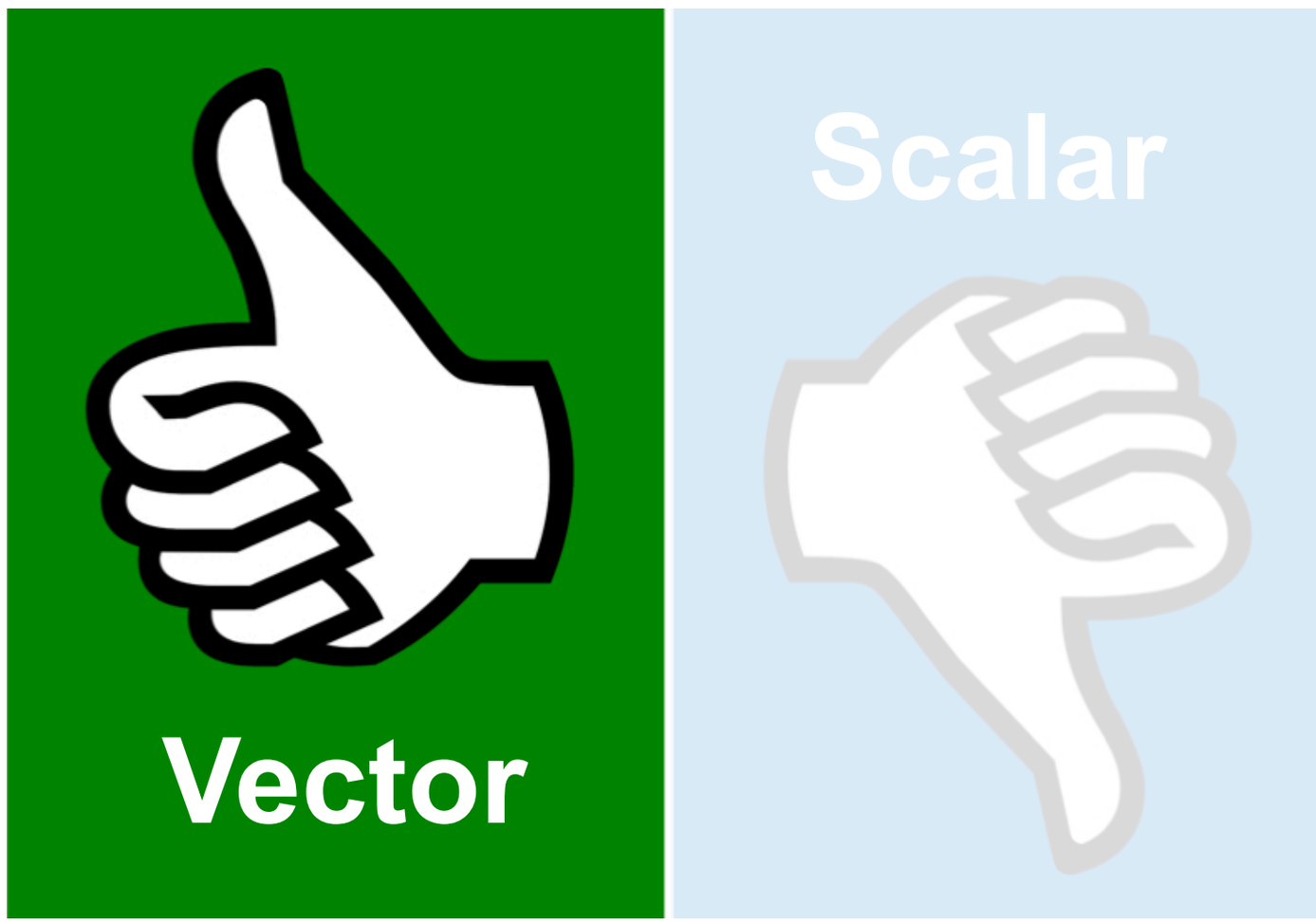
Electric Forces

§ 5.3
Lecture 02

Youtube Playlist

Knowledge Review:
Forces

A force is a…



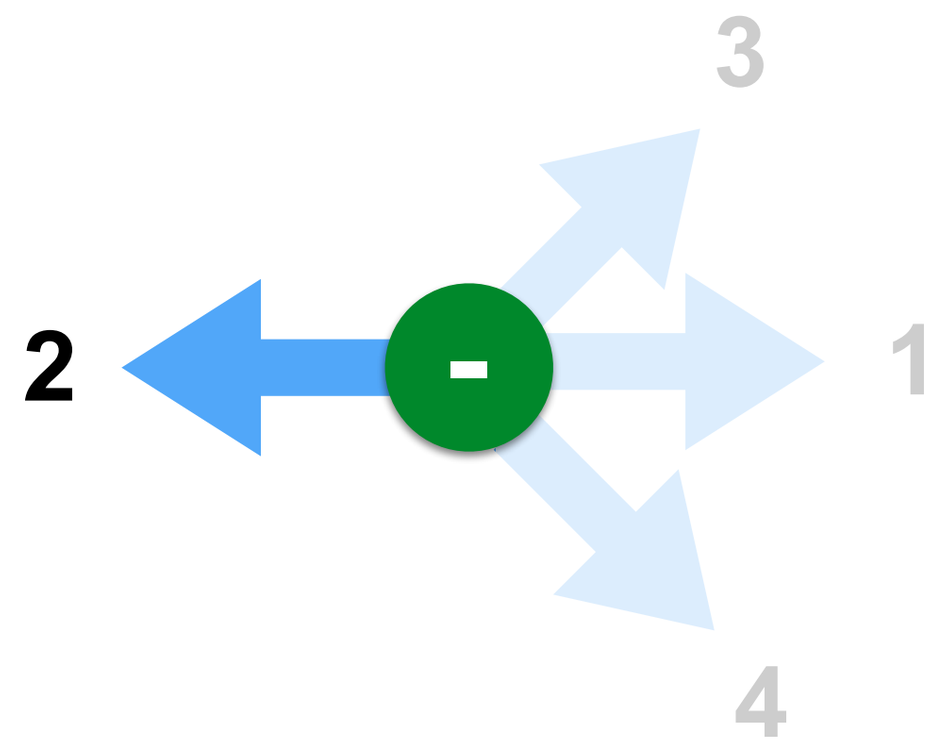
0. Don’t wait on my response

Charged Balls

Electric Forces

These two near by balls are electrically charged as shown below.

Which arrow shows the direction of the *electric force* on the negatively charged ball?



0. Don’t wait on my response

Electric Forces

Positive charges : Repel other positive charges
Attract negative charges

Negative charges : Attract positive charges
Repel other negative charges

Neutral charges : Feel no electric force

Electric Hockey

Coulomb's Law

q_1

q_2

Electrostatic constant
$K = 8.99 \times 10^9$

(*Magnitude* of) each objects' charge

$$F_{1\text{ on }2} = F_{2\text{ on }1} = K\frac{|q_1|\ |q_2|}{r^2}$$

***Magnitude* of Electric Forces**

Equal because of Newton's Third Law

Distance between the two charges

Unit Check

Electrostatic Constant

To get a force in Newtons, what units must K have?

1. N m
2. N m / C
3. N m / C^2
4. N m^2 / C^2
5. N C^2 / m^2
0. Don't wait on my response

Coulombs, C

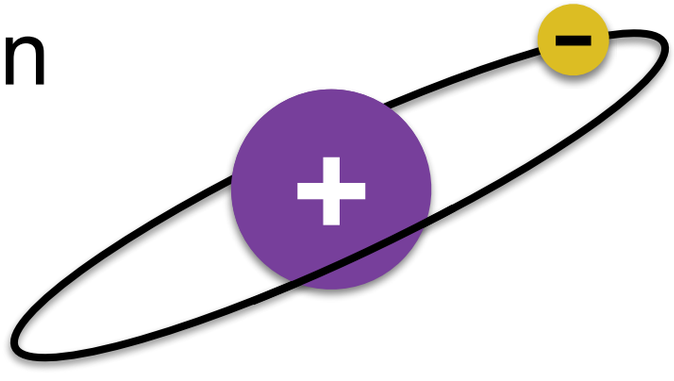
$$F = K \frac{|q_1| \; |q_2|}{r^2}$$

meters, m

$$\frac{N \cancel{m^2}}{\cancel{C^2}} \; \frac{\cancel{C} \; \cancel{C}}{\cancel{m^2}} \longrightarrow N$$

Hydrogen Atom

A hydrogen atom is made of one electron orbiting around a proton. The radius of the hydrogen atom is 5.29×10^{-11} m.



What is the force (magnitude and direction) that the electron feels?

$$F = k \frac{|q_{electron}| \; |q_{proton}|}{r^2}$$

$$F = 8.99\times10^{9} \frac{(1.60\times10^{-19})\,(1.60\times10^{-19})}{(5.29 \times10^{-11})^2}$$

$\vec{F} = 8.22\times10^{-8}$ N = 82.2 nN, inward toward proton

Using Coulomb's Law

This form of Coulomb's Law gives you just the <u>magnitude</u> of the forces.

$$F = K \frac{|q_1| \; |q_2|}{r^2}$$

You figure out directions based on sign of the charges.

Knowing the magnitude and direction of force(s), you can use all the knowledge you have from <u>Newton's Laws</u>.